

幂级数 + Taylor级数 + 无穷限积分

一. 求 $\sum_{n=1}^{\infty} \left(\frac{1}{2n+1} - 1\right) x^{2n}$ 在 $(-1, 1)$ 内的和函数 $S(x)$

解: $\sum_{n=1}^{\infty} \frac{x^{2n}}{2n+1} - \sum_{n=1}^{\infty} x^{2n}$

$$= \frac{1}{x} \sum_{n=1}^{\infty} \frac{x^{2n+1}}{2n+1} - x \sum_{n=0}^{\infty} (x^2)^n$$

$$= \frac{1}{x} \sum_{n=1}^{\infty} \frac{x^{2n+1}}{2n+1} - \frac{x^2}{1-x^2}$$

$$\left(\sum_{n=1}^{\infty} \frac{x^{2n+1}}{2n+1} \right)' = \sum_{n=1}^{\infty} x^{2n} = x^2 \sum_{n=0}^{\infty} x^{2n} = \frac{x^2}{1-x^2}$$

$$\therefore \sum_{n=1}^{\infty} \frac{x^{2n+1}}{2n+1} - 0 = \int_0^x \frac{t^2}{1-t^2} dt = \int_0^x \left(-1 + \frac{1}{1-t^2}\right) dt$$

$$= -t \Big|_0^x + \int_0^x \frac{1}{1-t^2} dt$$

$$= -x + \frac{1}{2} \int_0^x \left(\frac{1}{1-t} - \frac{1}{1+t} \right) dt$$

$$= -x + \frac{1}{2} \int_0^x \frac{1}{1-t} dt - \frac{1}{2} \int_0^x \frac{1}{1+t} dt$$

$$= -x - \frac{1}{2} \int_0^x \frac{1}{1-t} d(1-t) - \frac{1}{2} \int_0^x \frac{1}{1+t} d(1+t)$$

$$= -x - \frac{1}{2} \ln|1-t| \Big|_0^x - \frac{1}{2} \ln|1+t| \Big|_0^x$$

$$= -x + \frac{1}{2} \ln|1-x| - \frac{1}{2} \ln|1+x|, x \in (-1, 1)$$

$$= -x + \frac{1}{2} \ln(1-x) - \frac{1}{2} \ln(1+x)$$

$$= -x + \frac{1}{2} \ln \frac{1+x}{1-x}, |x| \in (0, 1)$$

$$f_1(x) = -1 + \frac{1}{2x} \ln \frac{1+x}{1-x} - \frac{x^2}{1-x^2}$$

$$= \frac{1}{2x} \ln \frac{1+x}{1-x} - \frac{x^2+x^2}{1-x^2} = \frac{1}{2x} \ln \frac{1+x}{1-x} - \frac{1}{1-x^2}, |x| \in (0, 1)$$

当 $x=0$ 时, $S=0$

$$\therefore S(x) = \begin{cases} 0, & x=0 \\ \frac{1}{2x} \ln \frac{1+x}{1-x} - \frac{1}{1-x^2}, & 0 < |x| < 1 \end{cases}$$

二. 求 $\sum_{n=1}^{\infty} (-1)^{n-1} \left(1 + \frac{1}{n(2n-1)}\right) x^{2n}$ 的收敛域与和函数 $f(x)$

解: $\lim_{n \rightarrow \infty} \left| \frac{U_{n+1}}{U_n} \right| = \lim_{n \rightarrow \infty} \frac{\left(1 + \frac{1}{(n+1)(2n+1)}\right) x^{2n+2}}{\left(1 + \frac{1}{n(2n-1)}\right) x^{2n}} = x^2 < 1$

$x \in (-1, 1)$ 收敛收敛

$\therefore$ 收敛域为 $(-1, 1)$

$$f(x) = \sum_{n=1}^{\infty} (-1)^{n-1} x^{2n} + \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n(2n-1)} x^{2n}$$

$$\frac{1}{2} f_1(x) = \sum_{n=1}^{\infty} (-1)^{n-1} x^{2n} = - \sum_{n=1}^{\infty} (-1)^n (x^2)^n$$

$$= - \sum_{n=1}^{\infty} (-x^2)^n = -(x^2) \sum_{n=0}^{\infty} (-x^2)^n$$

$$= x^2 \frac{1}{1+x^2} = \frac{x^2}{1+x^2}$$

$$\frac{1}{2} f_2(x) = \sum_{n=1}^{\infty} (-1)^n \frac{1}{2n(2n-1)} x^{2n}$$

$$f_2'(x) = 2 \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{2n-1}$$

$$f_2''(x) = 2 \sum_{n=1}^{\infty} (-1)^n x^{2n-2} = 2 \sum_{n=0}^{\infty} (-x^2)^n = \frac{2}{1+x^2}$$

$$f_2'(0) = 0, f_2(0) = 0$$

$$\therefore f_2'(x) = \int_0^x f_2''(t) dt = 2 \arctan x$$

$$f_2(x) = \int_0^x f_2'(t) dt = 2 \int_0^x \arctan t dt$$

$$= 2 \arctan t \Big|_0^x - 2 \int_0^x \frac{t}{t^2+1} dt$$

$$= 2 \arctan x - \int_0^x \frac{1}{t^2+1} d(t^2+1)$$

$$= 2 \arctan x - \ln(x^2+1)$$

$$= f(x) = f_1(x) + f_2(x)$$

$$\Rightarrow f(x) = \frac{x^2}{x^2+1} + 2 \arctan x - \ln(x^2+1)$$

三 求级数 $\sum_{n=0}^{\infty} (-1)^n \frac{1}{2^n} C_{2n}^{2n-1} \log 2$

解: $f(x) = f(x) = \sum_{n=0}^{\infty} C_{2n}^{2n-1} \frac{(-1)^n}{2^n} + \sum_{n=0}^{\infty} (-1)^n$

$= f_1(x) + f_2(x)$

$f_2(x) = \sum_{n=0}^{\infty} (-\frac{1}{2})^n = \frac{1}{1+\frac{1}{2}} = \frac{2}{3}$

求 $f_1(x)$ 即

$f(x) = \sum_{n=0}^{\infty} C_{2n}^{2n-1} (-\frac{1}{2})^n = \sum_{n=0}^{\infty} n(n-1) (-\frac{1}{2})^n$

$= \frac{1}{4} \sum_{n=0}^{\infty} n(n-1) (-\frac{1}{2})^{n-2} = \frac{1}{4} \left[\sum_{n=0}^{\infty} (-\frac{1}{2})^n \right]''$

$\sum_{n=0}^{\infty} (-\frac{1}{2})^n \Rightarrow \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$

$(\frac{1}{1-x})' = \frac{-(-1)}{(1-x)^2} = \frac{1}{(1-x)^2}$

$(\frac{1}{1-x})'' = -\frac{-2+2x}{(1-x)^3} = -\frac{2}{(1-x)^3} = -\frac{2}{(-\frac{1}{2}-1)^3} = \frac{6}{27}$

$= f_1(x) \cdot \frac{1}{4} \times \frac{6}{27} = \frac{4}{27}$

$f(x) = f_1(x) + f_2(x) = \frac{4}{27} + \frac{2}{3} = \frac{22}{27}$

四. 求 $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \cos x dx, n=0,1,2,\dots$ 并 $\sum_{n=0}^{\infty} I_n$

解: $I_n = \int_0^{\frac{\pi}{2}} \sin^n x d(\sin x)$

$\Rightarrow \frac{\pi}{2} = \sin x \in (0, \frac{\sqrt{2}}{2}, 1]$

$I_n = \int_0^{\frac{\sqrt{2}}{2}} t^n dt = \frac{1}{n+1} t^{n+1} \Big|_0^{\frac{\sqrt{2}}{2}}$

$I_n = \frac{1}{n+1} (\frac{1}{\sqrt{2}})^{n+1}$

$\therefore \sum_{n=0}^{\infty} I_n = \sum_{n=0}^{\infty} \frac{1}{n+1} (\frac{1}{\sqrt{2}})^{n+1}$

$\frac{1}{2} f(x) = \sum_{n=0}^{\infty} \frac{1}{n+1} x^{n+1}$

$f(x) = \sum_{n=0}^{\infty} x^{n+1} = \frac{1}{1-x}, f(0) = 0$

$f(x) = \int_0^x \frac{1}{1-t} dt = -\int_0^x \frac{1}{1-t} d(1-t)$
 $= -\ln|1-t| \Big|_0^x$

$= -\ln|1-x|$

$\therefore x = \frac{1}{\sqrt{2}} = f(x) = -\ln(1-\frac{1}{\sqrt{2}})$

$= f(x) = -\ln(1-\frac{1}{\sqrt{2}}) = \ln \frac{\sqrt{2}}{\sqrt{2}-1}$

$= \ln \frac{\sqrt{2}(\sqrt{2}+1)}{2-1} = \ln(2+\sqrt{2})$

五 已知 $f_n(x)$ 满足 $f_n'(x) = f_n(x) + x^{n-1} e^x$ (n 为正整数), $f_0(x) = \frac{e}{x}$.

求 $\sum_{n=1}^{\infty} f_n(x)$ 及 f_n

解: $f_n'(x) = f_n(x) + x^{n-1} e^x$

$$f_n'(x) - f_n(x) = x^{n-1} e^x$$

$$\Rightarrow f_n(x) = e^x \left(\frac{x^n}{n} + C \right)$$

$$\because f_n(1) = \frac{e}{n} \Rightarrow e \left(\frac{1}{n} + C \right) = \frac{e}{n} \Rightarrow C = 0$$

$$\therefore f_n(x) = \frac{e^x x^n}{n}$$

即 $\sum_{n=1}^{\infty} \frac{e^x x^n}{n}$ 收敛

$$\sum_{n=1}^{\infty} \frac{e^x x^n}{n} = e^x \sum_{n=1}^{\infty} \frac{x^n}{n}$$

$$\text{即 } f(x) = \sum_{n=1}^{\infty} \frac{x^n}{n}$$

$$f'(x) = \sum_{n=1}^{\infty} x^{n-1} = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}, \quad f(0) = 0$$

$$f(x) - f(0) = \int_0^x \frac{1}{1-t} dt$$

$$= - \int_0^x \frac{1}{1-t} d(1-t)$$

$$= - \ln(1-t) \Big|_0^x$$

$$= - \ln(1-x) = - \ln(1-x)$$

$$f(x) = - \ln(1-x)$$

$$\therefore \sum_{n=1}^{\infty} \frac{e^x x^n}{n} = e^x \times (- \ln(1-x)) = - e^x \ln(1-x), \quad x \in (-1, 1)$$

六. 用间接法求将函数展开成幂级数

(1) $f(x) = e^{-x^2}$

解: 令 $-x^2 = t$

$$f(t) = e^t = \sum_{n=0}^{\infty} \frac{t^n}{n!}$$

$$\text{故 } f(x) = \sum_{n=0}^{\infty} \frac{(-x^2)^n}{n!} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{n!}, \quad x \in \mathbb{R}$$

(2) $f(x) = \sinh x$

$$\frac{1}{2} f(x) = \sinh t, \quad t = 2x, \quad |t|$$

$$\sinh t = \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n+1}}{(2n+1)!} \quad \text{故 } f(x) = \sum_{n=0}^{\infty} (-1)^n \frac{(2x)^{2n+1}}{(2n+1)!}$$

(3) $f(x) = \sinh^2 x$

$$\text{解: } f(x) = \sinh^2 x = \frac{1 - \cosh 2x}{2} = \frac{1}{2} - \frac{1}{2} \cosh 2x = \frac{1}{2} - \frac{1}{2} \cosh t, \quad (t = 2x)$$

$$(1 - 2 \sinh^2 x = \cosh 2x)$$

$$\sinh t = \sum_{n=0}^{\infty} (-1)^n \frac{t^{2n+1}}{(2n+1)!}$$

$$\cosh t = (\sinh t)' = \sum_{n=0}^{\infty} (-1)^n \frac{t^{2n}}{(2n)!}$$

$$\text{故 } f(x) = \frac{1}{2} - \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \frac{(2x)^{2n}}{(2n)!}, \quad x \in \mathbb{R}$$

(4) $f(x) = \arctan x$

解: $f(x) = \arctan x$

$$f'(x) = \frac{1}{1+x^2} = \frac{1}{1-x^2} = \sum_{n=0}^{\infty} (-x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n}$$

$$x \in (-1, 1)$$

$$\begin{aligned} \arctan x = f(x) &= \int_0^x \frac{1}{1+t^2} dt = \sum_{n=0}^{\infty} \int_0^x (-1)^n t^{2n} dt \\ &= \sum_{n=0}^{\infty} (-1)^n \int_0^x t^{2n} dt = \sum_{n=0}^{\infty} (-1)^n \frac{t^{2n+1}}{2n+1} \Big|_0^x \\ &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \end{aligned}$$

$$5) f(x) = \frac{1}{5-2x}$$

$$\text{Ans: } f(x) = \frac{1}{5} \frac{1}{1 - (\frac{2}{5}x)}$$

$$\text{Let } \frac{2}{5}x = t, \text{ then } f(t) = \frac{1}{5} \frac{1}{1-t}$$

$$= \frac{1}{5} \sum_{n=0}^{\infty} t^n, \quad t \in (-1, 1)$$

$$\frac{2}{5}x \in (-1, 1), \quad x \in (-\frac{5}{2}, \frac{5}{2})$$

$$\text{Ans: } \frac{1}{5} \sum_{n=0}^{\infty} \left(\frac{2}{5}x\right)^n, \quad x \in (-\frac{5}{2}, \frac{5}{2})$$

$$6. f(x) = \frac{x}{9+x^2}$$

$$\text{Ans: } f(x) = \frac{x}{9} \frac{1}{1 + (\frac{x}{3})^2} = \frac{x}{9} \frac{1}{1 - [-(\frac{x}{3})^2]} = \frac{x}{9} \sum_{n=0}^{\infty} \left(-\frac{x^2}{9}\right)^n, \quad x \in (-3, 3)$$

$$\Rightarrow f(x) = \frac{x}{9} \sum_{n=0}^{\infty} \left(-\frac{1}{9}\right)^n x^{2n} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{3^{2n+2}}, \quad x \in (-3, 3)$$

$$7. f(x) = \ln(1+x+x^2+\dots+x^5)$$

$$\text{Ans: } f(x) = \ln\left(\frac{1-x^6}{1-x}\right) = \ln(1-x^6) - \ln(1-x), \quad x \neq 1$$

$$[\ln(1-x)]' = \frac{-1}{1-x} = -\sum_{n=0}^{\infty} x^n$$

$$\ln(1-x) = -\sum_{n=0}^{\infty} \int_0^x t^n dt = -\sum_{n=0}^{\infty} \frac{t^{n+1}}{n+1} \Big|_0^x$$

$$= -\sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1} = -\sum_{n=1}^{\infty} \frac{x^n}{n}$$

$$\text{Ans: } \ln(1-x^6) = -\sum_{n=1}^{\infty} \frac{x^{6n}}{n} = -\sum_{n=1}^{\infty} \frac{x^{6n}}{n}$$

$$\text{Ans: } \sum_{n=1}^{\infty} \frac{x^n}{n} - \sum_{n=1}^{\infty} \frac{x^{6n}}{n}, \quad x \in (-1, 1)$$

$$8) f(x) = \ln(x + \sqrt{1+x^2})$$

$$\text{Ans: } f(x) = \frac{1 + \frac{x}{\sqrt{1+x^2}}}{x + \sqrt{1+x^2}} = \frac{1}{\sqrt{1+x^2}}$$

$$\frac{1}{\sqrt{1+t}} = 1 + \sum_{n=1}^{\infty} (-1)^n \frac{(2n-1)!!}{(2n)!!} t^n, \quad -1 < t \leq 1$$

$$f(x) = \frac{1 + \frac{x}{\sqrt{1+x^2}}}{x + \sqrt{1+x^2}} = 1 + \sum_{n=1}^{\infty} (-1)^n \frac{(2n-1)!!}{(2n)!!} x^{2n}, \quad |x| \leq 1$$

$$\Rightarrow \text{Ans: } \int_0^x \frac{1}{\sqrt{1+t^2}} dt = x + \sum_{n=1}^{\infty} (-1)^n \frac{(2n-1)!!}{(2n)!!(2n+1)!!} \frac{x^{2n+1}}{2n+1}, \quad |x| \leq 1$$

$$9. f(x) = \ln\left(\frac{1+x}{1-x}\right)$$

$$\text{Ans: } f(x) = \frac{1+x}{1-x} \cdot \frac{1-x+1+x}{(1-x)^2} = \frac{2}{(1-x)^2} \frac{1+x}{1-x}$$

$$= 2 \frac{1}{1-(x^2)} = 2 \sum_{n=0}^{\infty} (x^2)^n$$

$$= 2 \sum_{n=0}^{\infty} x^{2n}$$

$$f(x) = 2 \sum_{n=0}^{\infty} \int_0^x t^{2n} dt$$

$$= 2 \sum_{n=0}^{\infty} \frac{t^{2n+1}}{2n+1} \Big|_0^x$$

$$= 2 \sum_{n=0}^{\infty} \frac{x^{2n+1}}{2n+1}, \quad x \in (-1, 1)$$

$$f(x) = 2 \sum_{n=0}^{\infty} \frac{x^{2n+1}}{2n+1}, \quad -1 < x < 1$$

10. $f(x) = \frac{1}{2} \ln \frac{1+x}{1-x} + \frac{1}{2} \arctan x - x.$

Ans: $f(x) = \frac{1}{2} \ln \frac{1+x}{1-x} + \frac{1}{2(1+x^2)} - 1$
 $= \frac{1}{2(1+x)(1-x)} + \frac{1}{2(1+x^2)} - 1$
 $= \frac{1+x^2+1-x^2}{2(1+x^2)(1-x^2)} - 1$

$= \frac{1}{1-x^4} - 1 = \sum_{n=0}^{\infty} x^{4n} - 1 = \sum_{n=1}^{\infty} x^{4n}$

$f(0) = 0$

$f(x) = \int_0^x \sum_{n=1}^{\infty} 4n x^{4n-1} dx = \sum_{n=1}^{\infty} \frac{4n}{4n+1} x^{4n+1} \Big|_0^x - x$

$f(x) = \sum_{n=1}^{\infty} \frac{x^{4n+1}}{n(n+1)}, x \in (-1, 1)$

七. 展成幂级数

1. $f(x) = \ln x, x = 2$

$\ln(t+1) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{t^n}{n}, t \in (-1, 1]$

$\ln x = \ln(x-2+2) = \ln 2 + \ln\left(\frac{x-2}{2} + 1\right)$

$\frac{x-2}{2} \in (-1, 1] \Rightarrow x \in (0, 4]$

Ans: $\ln 2 + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(x-2)^n}{n \cdot 2^n}, x \in (0, 4]$

(2) $f(x) = e^x, x = 1$

$f(x) = e^x = e e^{x-1} = e \cdot \sum_{n=0}^{\infty} \frac{(x-1)^n}{n!}, x \in \mathbb{R}$

(2) $f(x) = \frac{1}{x^2+2}, x = 1$

$f(x) = \frac{1}{(x+1)(x+2)} = \frac{1}{x+1} - \frac{1}{x+2}$

$= \frac{1}{2(1+x)} - \frac{1}{2(1+x+1)}$

$= \frac{1}{2} \frac{1}{1+\frac{x-1}{2}} - \frac{1}{2} \frac{1}{1+\frac{x+1}{2}}$

$= \sum_{n=0}^{\infty} (-1)^n \frac{(x-1)^n}{2^{n+1}} + \sum_{n=0}^{\infty} (-1)^{n+1} \frac{(x+1)^n}{2^{n+1}}, n \in \mathbb{N}$

$\Rightarrow f(x) = \sum_{n=0}^{\infty} (-1)^n \left(\frac{1}{2^{n+1}} - \frac{1}{2^{n+1}} \right) (x-1)^n, x \in \mathbb{R}$

(4) $f(x) = \sinh x, x = \frac{2}{3}$

$f(x) = \sinh x = \sinh\left(x - \frac{2}{3} + \frac{2}{3}\right) = \frac{e^{\frac{2}{3}}}{2} \left[\cosh\left(x - \frac{2}{3}\right) + \sinh\left(x - \frac{2}{3}\right) \right]$

$\cosh\left(x - \frac{2}{3}\right) = \sum_{n=0}^{\infty} (-1)^n \frac{\left(x - \frac{2}{3}\right)^{2n}}{(2n)!}, x \in (-\infty, +\infty)$

$\sinh\left(x - \frac{2}{3}\right) = \sum_{n=0}^{\infty} (-1)^n \frac{\left(x - \frac{2}{3}\right)^{2n+1}}{(2n+1)!}$

$\therefore f(x) = \frac{e^{\frac{2}{3}}}{2} \sum_{n=0}^{\infty} (-1)^n \left[\frac{\left(x - \frac{2}{3}\right)^{2n}}{(2n)!} + \frac{\left(x - \frac{2}{3}\right)^{2n+1}}{(2n+1)!} \right], x \in (-\infty, +\infty)$

1. 求函数 $f(x) = x^2 \ln(1+x)$ 在 $x=0$ 处 n 阶导数 $f^{(n)}(0)$

解: $f(x) = x^2 \ln(1+x)$

$$\ln(1+x) \text{ 展成级数 } \ln(1+x) = \sum_{k=0}^{\infty} (-1)^{k-1} \frac{x^k}{k}$$

$$f(x) = \sum_{k=0}^{\infty} (-1)^{k-1} \frac{x^{k+2}}{k}$$

$$f^{(n)}(x) = \sum_{k=n-2}^{\infty} (-1)^{k-1} \frac{(k+2)(k+1) \cdots (k+3-n)}{k} x^{k+2-n}$$

$f^{(n)}(0) = \text{求 } n \text{ 阶导数, 则 } k=n-2$

$$f^{(n)}(0) = (-1)^{n-3} \frac{n!}{n-2} = (-1)^{n-3} n(n-1)(n-2)!$$

(3) $\int_0^{+\infty} e^{-t} \sin wt \, dt$

$$\text{解: } \int_0^{+\infty} e^{-t} \sin wt \, dt = -\frac{1}{w} e^{-t} \cos wt \Big|_0^{+\infty} + \frac{1}{w} \int_0^{+\infty} e^{-t} \cos wt \, dt$$

$$= \frac{1}{w} - \frac{1}{w^2} \int_0^{+\infty} e^{-t} \cos wt \, dt$$

$$= \frac{1}{w} - \frac{1}{w^2} e^{-t} \sin wt \Big|_0^{+\infty} + \frac{1}{w^2} \int_0^{+\infty} e^{-t} \sin wt \, dt$$

$$= \frac{1}{w} - \frac{1}{w^2} \int_0^{+\infty} e^{-t} \sin wt \, dt$$

$$\therefore \int_0^{+\infty} e^{-t} \sin wt \, dt = \frac{w}{w^2 + 1}$$

九. 计算反常积分

(1) $\int_0^{+\infty} \frac{dx}{e^x + e^{-x}}$

解: $\int_0^{+\infty} \frac{e^x dx}{e^{2x} + 1} = \int_0^{+\infty} \frac{d(e^x)}{(e^x)^2 + 1} = \arctan e^x \Big|_0^{+\infty}$
 $= \frac{\pi}{2} - 0 = \frac{\pi}{2}$

(2) $\int_1^{+\infty} \frac{\arctan x}{x^2} dx = -\frac{\arctan x}{x} \Big|_1^{+\infty} - \int_1^{+\infty} -\frac{1}{x} \cdot \frac{1}{x^2+1} dx$

$$= \frac{\arctan 1}{1} + \int_1^{+\infty} \frac{1}{x(x^2+1)} dx$$

$$= \frac{\pi}{4} + \int_1^{+\infty} \frac{x}{x^2(x^2+1)} dx$$

$$= \frac{\pi}{4} + \frac{1}{2} \int_1^{+\infty} \frac{1}{x(x^2+1)} d(x^2)$$

$$= \frac{\pi}{4} + \frac{1}{2} \int_1^{+\infty} \frac{1}{x^2} d(x^2) - \frac{1}{2} \int_1^{+\infty} \frac{1}{x^2+1} d(x^2)$$

$$= \frac{\pi}{4} + \frac{1}{2} \ln \frac{x^2}{x^2+1} \Big|_1^{+\infty} = \frac{\pi}{4} - \frac{1}{2} \ln \frac{1}{2} = \frac{\pi}{4} + \frac{1}{2} \ln 2$$

(3) $\int_{-\infty}^{+\infty} \frac{dx}{x^2+2}$

$$= \int_{-\infty}^{+\infty} \frac{dx}{(x+\sqrt{2})(x-\sqrt{2})} = \arctan(x+\sqrt{2}) \Big|_{-\infty}^{+\infty}$$

$$= \frac{\pi}{2} - (-\frac{\pi}{2}) = \pi$$

十 判断下列无穷级数的敛散性

$$(1) \int_0^{+\infty} x^\alpha e^{-x} dx.$$

取 $p=2, \alpha$:

$$\lim_{x \rightarrow +\infty} x^{\alpha+2} e^{-x}$$

$$= \lim_{x \rightarrow +\infty} \frac{x^{\alpha+2}}{e^x} = 0$$

$\therefore p > 1$.

$\therefore \int_0^{+\infty} x^\alpha e^{-x} dx$ 对于任何 α 均收敛.

$$(2) \int_0^{+\infty} \frac{x^2}{\sqrt{x^2+1}} dx.$$

$$\lim_{x \rightarrow +\infty} \frac{x^{2+p}}{(x^2+1)^{\frac{p}{2}}} dx$$

又由最高项论法知, 则:

$$p = \frac{5}{2} - 2 = \frac{1}{2}$$

$$\therefore \lim_{x \rightarrow +\infty} x^{\frac{1}{2}} \frac{x^2}{\sqrt{x^2+1}} = 1$$

$\therefore \int_0^{+\infty} \frac{x^2}{\sqrt{x^2+1}} dx$ 是发散的.